17.2: Transcription is the DNA-directed synthesis of RNA: *A Closer Look*

Now that we have considered the linguistic logic and evolutionary significance of the genetic code, we are ready to reexamine transcription, the first stage of gene expression, in greater detail.

Molecular Components of Transcription

Messenger RNA, the carrier of information from DNA to the cell's protein-synthesizing machinery, is transcribed from the template strand of a gene. An enzyme called an **RNA polymerase** pries the two strands of DNA apart and joins together RNA nucleotides complementary to the DNA template strand, thus elongating the RNA polynucleotide **(Figure 17.8)**. Like the DNA polymerases that function in DNA replication, RNA polymerases can assemble a polynucleotide only in its 5′→3′ direction, adding onto its 3′ end. Unlike DNA polymerases, however, RNA polymerases are able to start a chain from scratch; they don't need to add the first nucleotide onto a pre-existing primer.

Figure 17.8

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The stages of transcription: initiation, elongation, and termination.
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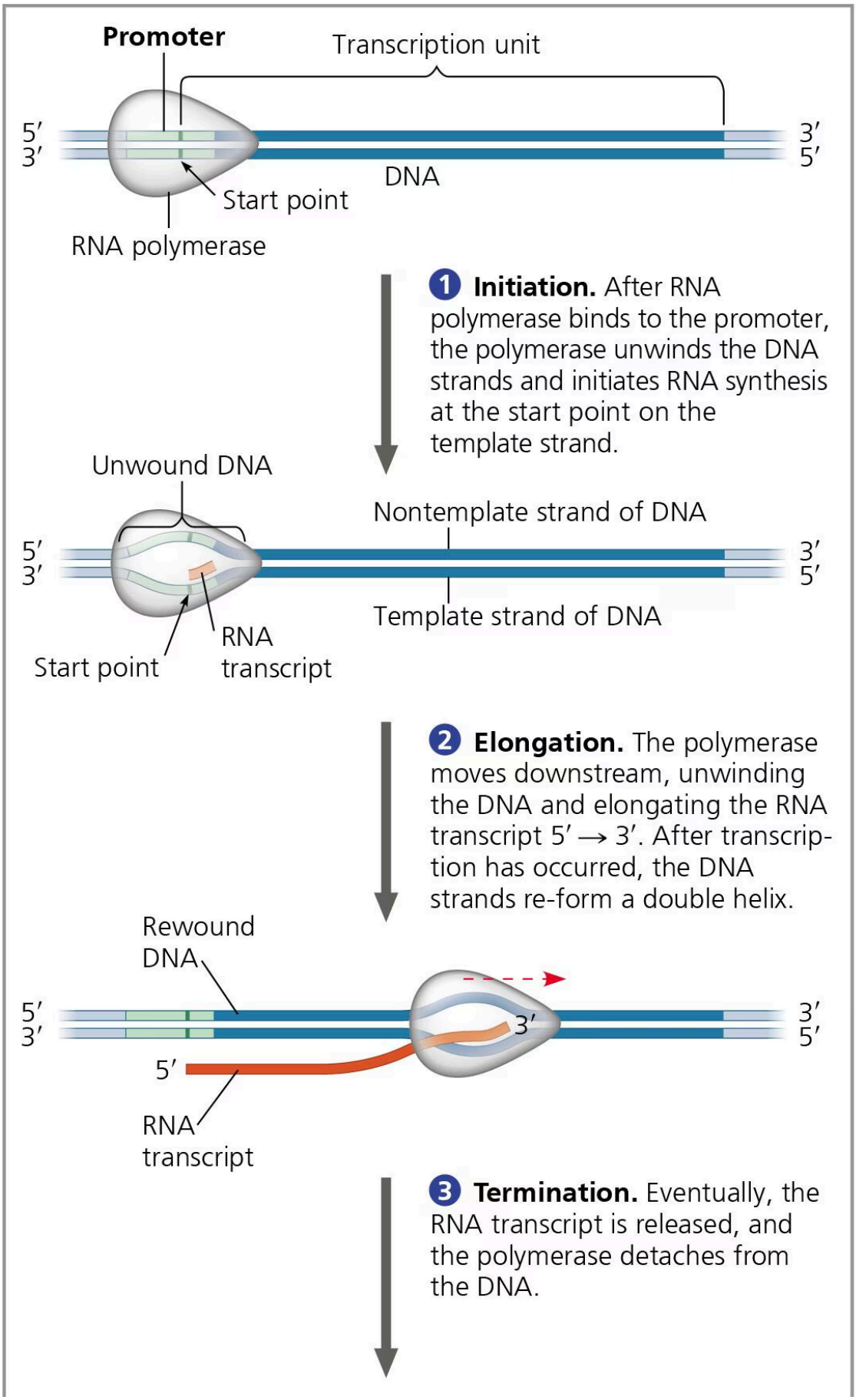
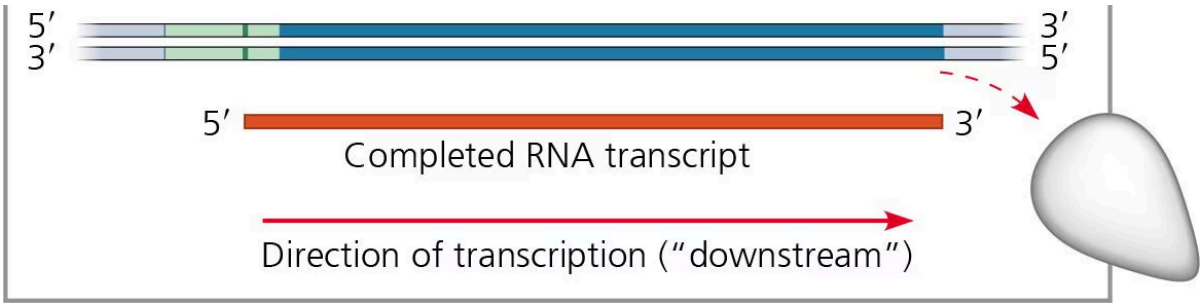

Promoter
Transcription unit
5′
3′
3′
5′
DNA
Start point
RNA polymerase
1 Initiation. After RNA polymerase binds to the promoter, the polymerase unwinds the DNA strands and initiates RNA synthesis at the start point on the template strand.
Unwound DNA
Nontemplate strand of DNA
5′
3′
3′
5′
Template strand of DNA
RNA transcript
Start point
2 Elongation. The polymerase moves downstream, unwinding the DNA and elongating the RNA transcript 5′ → 3′. After transcription has occurred, the DNA strands re-form a double helix.
Rewound DNA
5′
3′
3′
5′
3′
5′
RNA transcript
3 Termination. Eventually, the RNA transcript is released, and the polymerase detaches from the DNA.



This general depiction of transcription applies to both bacteria and eukaryotes, but the details of termination differ, as described in the text. Also, in a bacterium, the RNA transcript is immediately usable as mRNA; in a eukaryote, the RNA transcript must first undergo processing.

The stages are as follows.

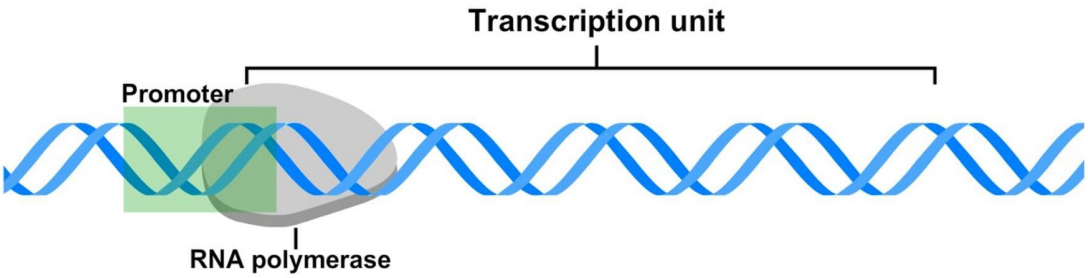
1. Initiation. After R N A polymerase binds to the promoter, D N A strands unwind, and polymerase initiates R N A synthesis at the start point on template strand for synthesis of the transcription unit. D N A template is the 3 prime to 5 prime strand; 5 prime to 3 prime strand is the nontemplate D N A. After initiation beings, a small strand of R N A transcript is visible within the R N A polymerase. Direction of transcription = downstream.
2. Elongation. The polymerase moves downstream, unwinding the D N A and elongating the R N A transcript 5 prime to 3 prime. A long strand of R N A is visible in this diagram. D N A winds back up after R N A polymerase passes. Direction of transcription is towards the 5 prime end of the D N A template.
3. Termination. Eventually, the R N A transcript is released, and the polymerase detaches from the D N A. D N A is rewound, and the R N A transcript is completed.

MAKE CONNECTIONS *Compare the use of a template strand during transcription and replication. See*Figure 16.18*.*

*For suggested answer, see*Appendix A*.*

Animation

Overview of Transcription



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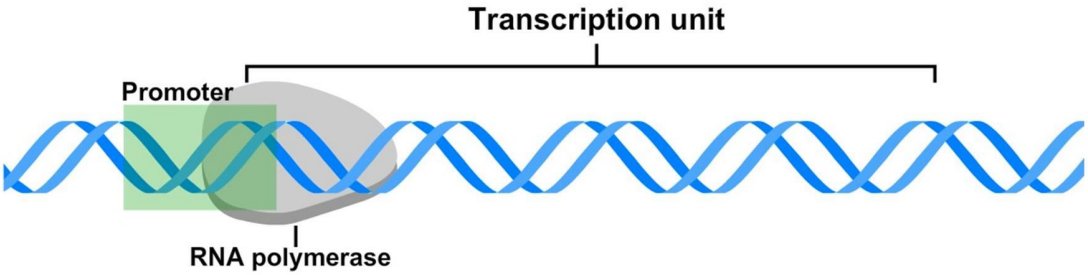

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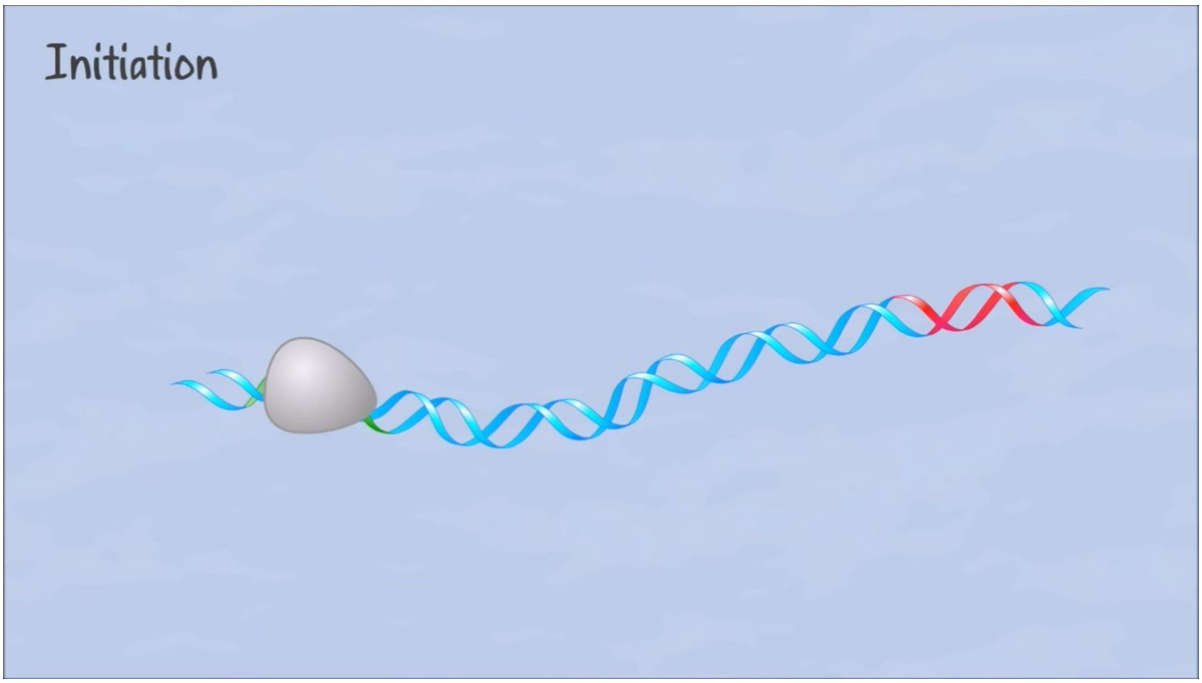
Animation

Overview of Transcription in Bacteria



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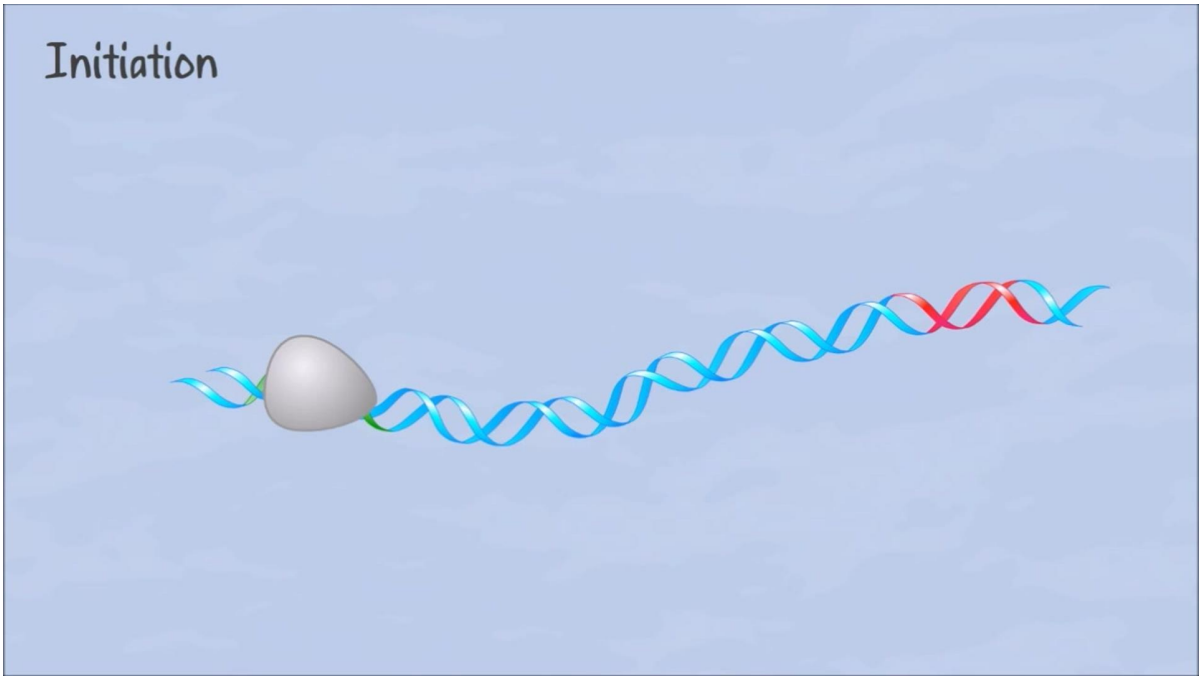

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Specific sequences of nucleotides along the DNA mark where transcription of a gene begins and ends. The DNA sequence where RNA polymerase attaches and initiates transcription is known as the **promoter**; in bacteria, the sequence that signals the end of transcription is called the **terminator**. (The termination mechanism in eukaryotes will be described later.) Molecular biologists refer to the direction of transcription

as “downstream” and the other direction as “upstream.” These terms are also used to describe the positions of nucleotide sequences within the DNA or RNA. Thus, the promoter sequence in DNA is said to be upstream from the terminator. The stretch of DNA downstream from the promoter that is transcribed into an RNA molecule is called a **transcription unit**.

Bacteria have a single type of RNA polymerase that synthesizes not only mRNA but also other types of RNA that function in gene expression, such as ribosomal RNA. In contrast, eukaryotes have at least three types of RNA polymerase in their nuclei; the one used for pre-mRNA synthesis is called RNA polymerase II. The other RNA polymerases transcribe RNA molecules that are not translated into protein. In the discussion that follows, we start with the features of mRNA synthesis common to both bacteria and eukaryotes and then describe some key differences.

Synthesis of an RNA Transcript

The three stages of transcription, as shown in Figure 17.8 and described next, are initiation, elongation, and termination of the RNA chain. Study Figure 17.8 to familiarize yourself with the stages and the terms used to describe them.

RNA Polymerase Binding and Initiation of Transcription

The promoter sequence of a gene includes within it the transcription **start point**—the nucleotide where RNA polymerase actually begins synthesizing the mRNA—and typically extends several dozen or so nucleotide pairs upstream from the start point **(Figure 17.9)**. Based on interactions with proteins (transcription factors), RNA polymerase binds in a precise location and orientation on the promoter. This binding determines where transcription starts and the direction it will travel, thus which strand of DNA is used as the template.

Figure 17.9

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The initiation of transcription at a eukaryotic promoter.
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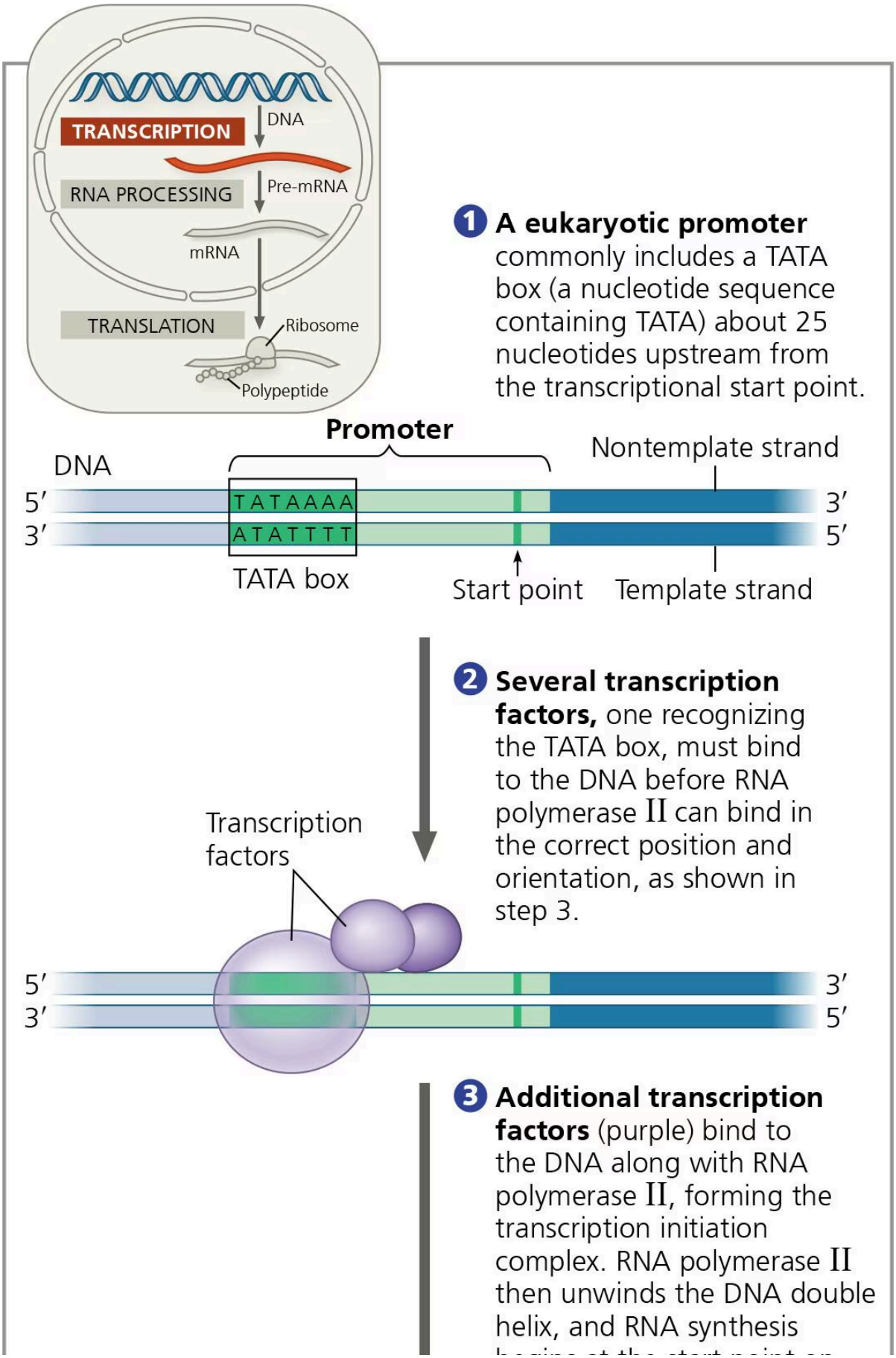
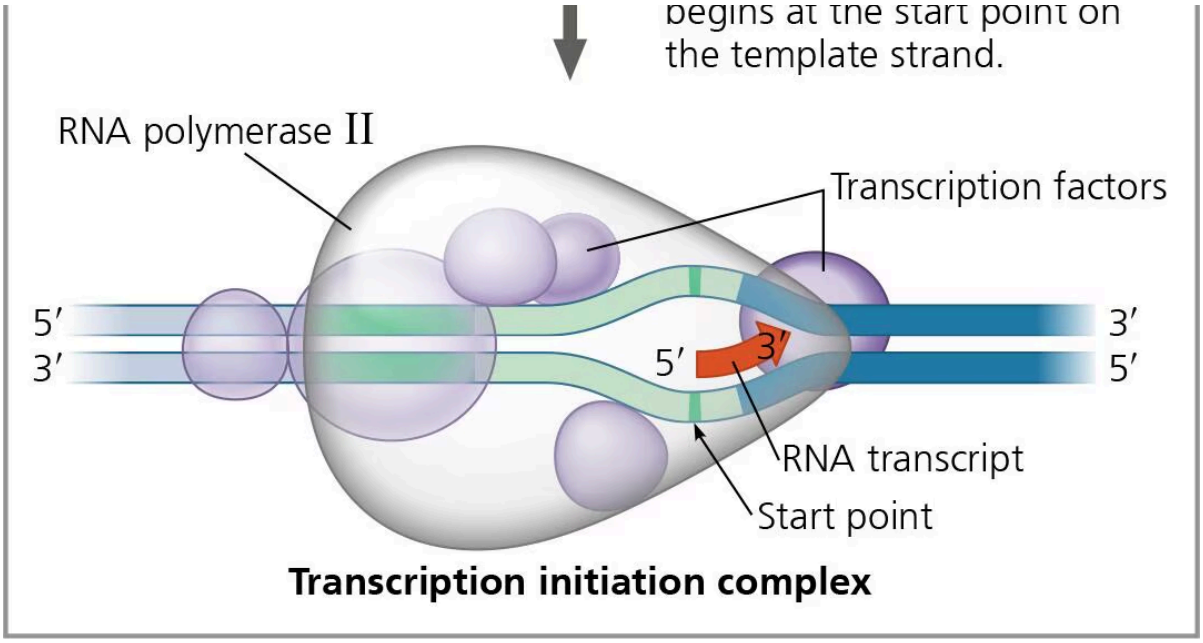

TRANSCRIPTION
DNA
RNA PROCESSING
Pre-mRNA
mRNA
TRANSLATION
Ribosome
Polypeptide
1 A eukaryotic promoter commonly includes a TATA box (a nucleotide sequence containing TATA) about 25 nucleotides upstream from the transcriptional start point.
Promoter
DNA
Nontemplate strand
5′
3′
TATAAAA
ATATTTT
3′
5′
TATA box
Start point
Template strand
2 Several transcription factors, one recognizing the TATA box, must bind to the DNA before RNA polymerase II can bind in the correct position and orientation, as shown in step 3.
Transcription factors
5′
3′
3′
5′
3 Additional transcription factors (purple) bind to the DNA along with RNA polymerase II, forming the transcription initiation complex. RNA polymerase II then unwinds the DNA double helix, and RNA synthesis



In eukaryotic cells, proteins called transcription factors mediate the initiation of transcription by RNA polymerase II.

The steps are as follows.

1. A eukaryotic promoter commonly includes a TATA box, a nucleotide sequence containing TATA, about 25 nucleotides upstream from the transcriptional start point. (By convention, nucleotide sequences are given as they occur on the nontemplate strand). Here, D N A drawn as top strand (5 prime to 3 prime, nontemplate) and bottom strand (3 prime to 5 prime, template. TATA box in the promoter region shows sequence T A T A A A A on the nontemplate strand. Corresponding sequence on the template strand is A T A T T T T. Start point is labeled on the template strand at the end of the promoter region.

2. Several transcription factors, one recognizing the TATA box, must bind to the D N A before R N A polymerase II can bind in the correct position and orientation. Here, transcription factors are drawn on the promoter region of the D N A near the TATA box.

3. Additional transcription factors bind to the D N A along with R N A polymerase II, forming the transcription initiation complex. R N A polymerase II then unwinds the D N A double helix, and R N A synthesis begins at the start point on the template strand. The R N A transcript is synthesized 5 prime to 3 prime on the D N A template strand (3 prime to 5 prime).

Explain how the interaction of RNA polymerase with the promoter would differ if the figure showed transcription initiation for bacteria.

*For suggested answer, see*Appendix A.**

Certain sections of a promoter sequence are especially important for binding RNA polymerase in a way that ensures that transcription will begin at the correct start point. In bacteria, part of the RNA polymerase itself specifically recognizes and binds to the promoter. In eukaryotes, a collection of proteins called **transcription factors** (purple proteins in Figure 17.9), help guide the binding of RNA polymerase and the initiation of transcription. Only after transcription factors are attached to the promoter does RNA polymerase II bind to it. The whole complex of transcription factors and RNA polymerase II bound to the promoter is called a **transcription initiation complex**. Figure 17.9 shows the role of transcription factors and a crucial promoter DNA sequence called the **TATA box** in forming the initiation complex at a eukaryotic promoter.

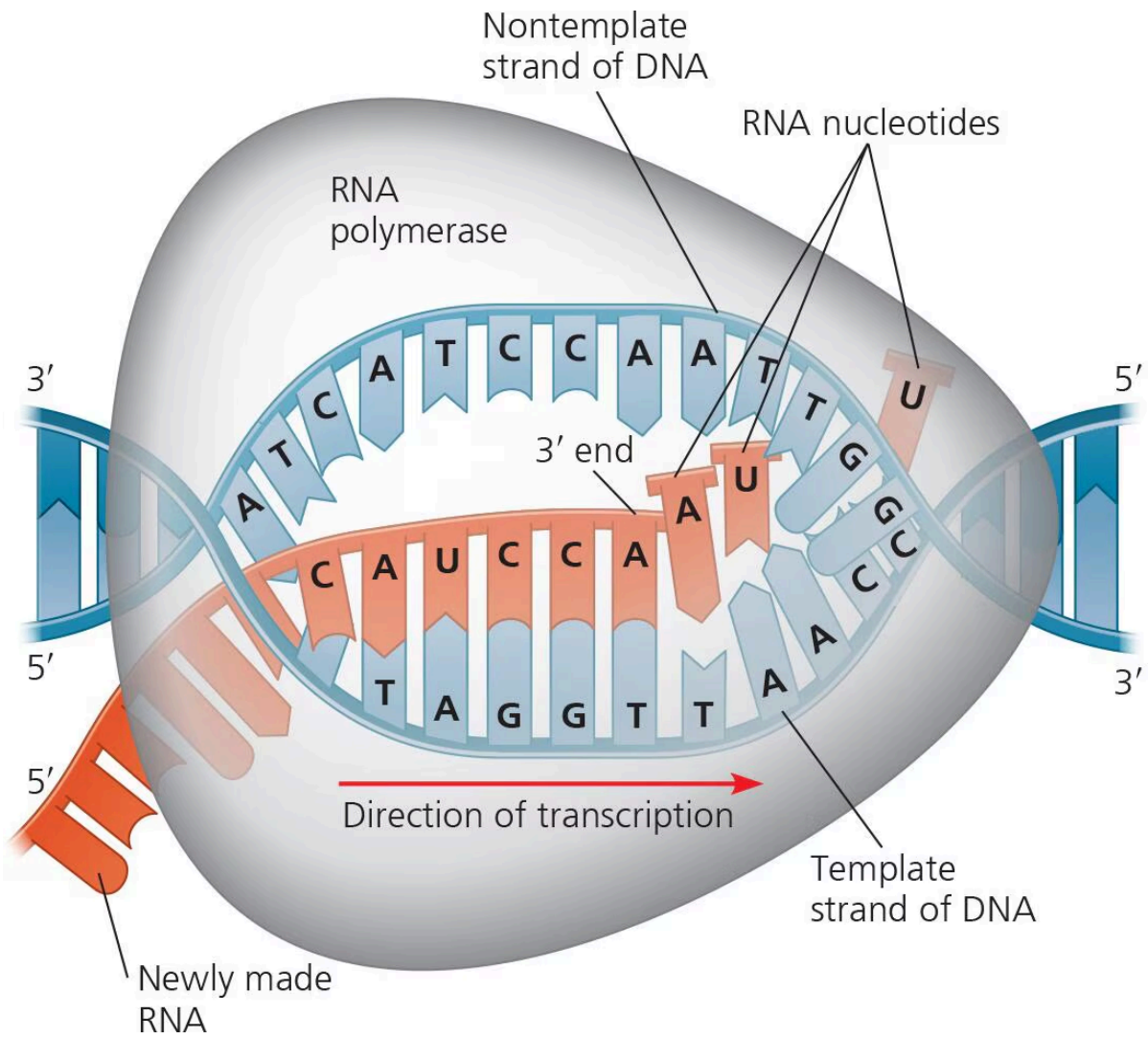
The interaction between eukaryotic RNA polymerase II and transcription factors is an example of the importance of protein-protein interactions in controlling eukaryotic transcription. Once the appropriate transcription factors are firmly attached to the promoter DNA and the polymerase is bound to them in the correct orientation on the DNA, the enzyme unwinds the two DNA strands and begins transcribing the template strand at the start point.

Elongation of the RNA Strand

As RNA polymerase moves along the DNA, it untwists the double helix, exposing about 10–20 DNA nucleotides at a time for pairing with RNA nucleotides **(Figure 17.10)**. The enzyme adds nucleotides to the 3′ end of the growing RNA molecule as it moves along the double helix. As transcription proceeds forward, the newly synthesized RNA molecule behind the RNA polymerase peels away from its DNA template, and the DNA double helix re-forms. Transcription progresses at a rate of about 40 nucleotides per second in eukaryotes.

Figure 17.10

Transcription elongation.



Template D N A nucleotide sequence is: 3 prime-T A G G T T A A-5 prime. R N A nucleotides added to template are: 5 prime-A U C C A-3 prime, and A and U are in the process of being added at the 3 prime end of the R N A strand.

RNA polymerase moves along the DNA template strand, joining complementary RNA nucleotides to the 3′ end of the growing RNA transcript. Behind the polymerase, the new RNA peels away from the template strand, which re-forms a double helix with the nontemplate strand.

BioFlix® Animation

Transcription

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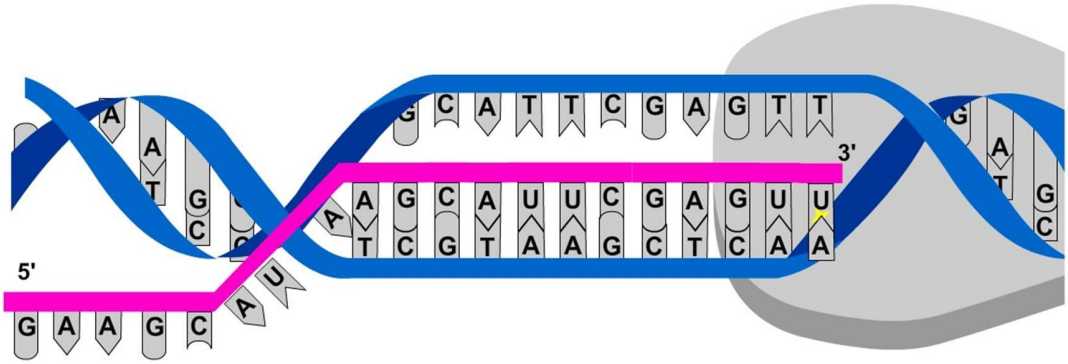
Animation

Elongation of the RNA Strand



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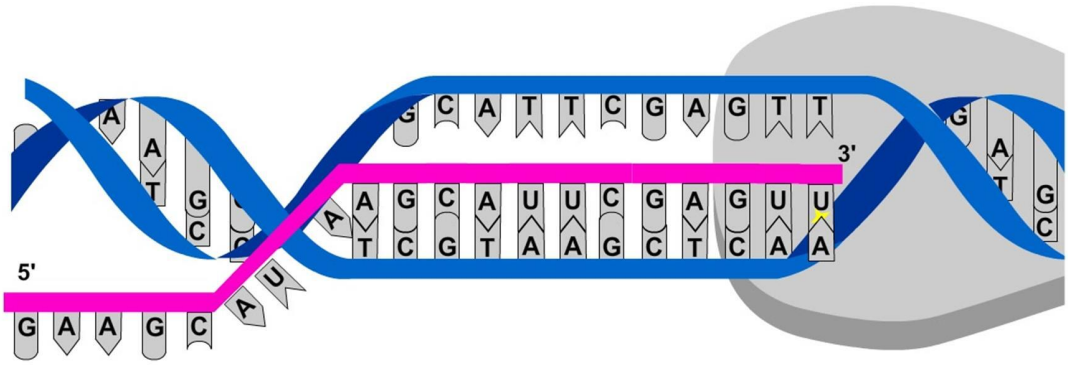

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A single gene can be transcribed simultaneously by several molecules of RNA polymerase following each other like trucks in a convoy. A growing strand of RNA trails off from each polymerase, with the length of each new strand reflecting how far along the template the enzyme has traveled from the start point (see the

mRNA molecules in Figure 17.23). The congregation of many polymerase molecules simultaneously transcribing a single gene increases the amount of mRNA transcribed from it, which helps the cell make the encoded protein in large amounts.

Termination of Transcription

Bacteria and eukaryotes differ in the way they terminate transcription. In bacteria, transcription proceeds through a terminator sequence in the DNA. The transcribed terminator (an RNA sequence) functions as the termination signal, causing the polymerase to detach from the DNA and release the transcript, which requires no further modification before translation. In eukaryotes, RNA polymerase II transcribes a sequence on the DNA called the polyadenylation signal sequence, which specifies a polyadenylation signal (AAUAAA) in the pre-mRNA. This is called a “signal” because once this stretch of six RNA nucleotides appears, it is immediately bound by certain proteins in the nucleus. Then, at a point about 10–35 nucleotides downstream from the AAUAAA, these proteins cut the RNA transcript free from the polymerase, releasing the pre-mRNA. The pre-mRNA then undergoes processing, the topic of the next section. Although that cleavage marks the end of the mRNA, the RNA polymerase II continues to transcribe. Enzymes begin to degrade the RNA made after cleavage, starting at its newly exposed 5′ end. The polymerase continues transcribing, pursued by the enzymes, until they catch up to the polymerase and it dissociates from the DNA.